KUZLAB - KUZAPP

- KUZAPP.md

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KuzApp - BETA - LINUX SYSTEM ADMINISTRATION TOOLS

GitHub KuzApp Bash PHP CSS HTML License Version

Apache (Web Host Version) + Apache (Docker Version) [KuzApp-Docker Branch]

Kuzapp-Web1 Kuzapp-Web1 Kuzapp-Web1 Kuzapp-Web1

Kuzapp-Web

OVERVIEW

KuzApp is a web-based application designed to provide Linux system administration tools through a user-friendly interface.
It allows users to execute various system administration scripts securely after logging in.
The application is currently in Beta version 0.2-2025 and supports features like user registration, secure login, and script execution for system updates, network checks, and more.

FEATURES

User Authentication --> Secure login and registration system with password hashing (SHA-256).

System Administration Tools --> Execute scripts for system updates, network checks, boot sequence analysis, and system information retrieval.

Responsive Design --> A visually appealing interface with a consistent theme using CSS styling.

SSL Support --> Runs on HTTPS with a self-signed certificate for secure communication.

PREREQUISITES

Operating System --> Ubuntu or Debian-based Linux distribution.

Software -->
Apache2 web server
PHP with MySQL support
MySQL/MariaDB database
OpenSSL for certificate generation

Network --> A static IP address (e.g., 192.168.124.187 as used in the configuration).

Git --> To clone the repository.

INSTALLATION INSTRUCTIONS --> 2 WAYS

1/ SCRIPT INSTALL

AUTOMATIK-INSTALL IN PROGRESS --> https://github.com/Kusanagi8200/KuzApp/tree/main/kuzapp-auto/kuzapp-auto-install.sh
AUTOMATIK-UNINSTALL IN PROGRESS --> https://github.com/Kusanagi8200/KuzApp/tree/main/kuzapp-auto/kuzapp-auto-uninstall.sh

Kuzapp-Web1

2/ ADMIN-SYS INSTALL

STEP 1 --> Install Required Packages

Run the following commands to install Apache2, PHP, and MySQL dependencies depends on your system

```
sudo apt update
sudo apt install apache2 php-mysql php-mysqli git php
sudo a2enmod ssl
sudo a2enmod rewrite
```

Verify that the mysqli PHP module is enabled

```
php -m | grep mysqli
```

STEP 2 --> Clone the KuzApp Repository

Clone the KuzApp repository from GitHub to the Apache web directory

```
sudo git clone https://github.com/Kusanagi8200/KuzApp.git /var/www/html/KuzApp
```

All necessary files are available in the repository --> https://github.com/Kusanagi8200/KuzApp/tree/main

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STEP 3 --> Set Up MySQL Database

Install MySQL/MariaDB and configure the database for KuzApp

```
sudo apt install mariadb-server
sudo mysql -u root -p
```

Create the database and user

```
CREATE DATABASE registration;
USE registration;

CREATE TABLE users (
id INT AUTO_INCREMENT PRIMARY KEY,
username VARCHAR(255) NOT NULL,
password VARCHAR(255) NOT NULL,
email VARCHAR(255) NOT NULL,
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE USER 'kuzapp_user'@'localhost' IDENTIFIED BY 'your_secure_password';
GRANT ALL PRIVILEGES ON registration.* TO 'kuzapp_user'@'localhost';
FLUSH PRIVILEGES;
EXIT;
```

Replace kuzapp_user and your_secure_password with your desired username and password.

STEP 4 --> Generate Self-Signed SSL Certificate

Create directories for SSL certificates and generate a self-signed certificate

```
sudo mkdir -p /etc/ssl/private /etc/ssl/certs

sudo openssl req -x509 -nodes -days 365 -newkey rsa:2048 \
-keyout /etc/ssl/private/kuzapp-selfsigned.key \
-out /etc/ssl/certs/kuzapp-selfsigned.crt \
-subj "/C=$COUNTRY/ST=$REGION/L=$CITY/O=KuzApp/OU=IT/CN=$IP"
```

Replace $IP with your server's IP address.

STEP 5 --> Configure Apache Virtual Host

Create a new Apache configuration file for KuzApp

```
sudo nano /etc/apache2/sites-available/KuzApp.conf
```

Add the following configuration

```
<VirtualHost *:8443>
ServerName $IP
DocumentRoot /var/www/html/KuzApp

SSLEngine on
SSLCertificateFile /etc/ssl/certs/kuzapp-selfsigned.crt
SSLCertificateKeyFile /etc/ssl/private/kuzapp-selfsigned.key

DirectoryIndex /KuzApp/login.php

<Directory /var/www/html/KuzApp>
    Require all granted
    AllowOverride All
    Options FollowSymLinks MultiViews
</Directory>
</VirtualHost>
```

Enable the site and restart Apache

```
sudo a2ensite KuzApp.conf
sudo systemctl restart apache2
```

STEP 6 --> Set Up File Permissions

Set the correct ownership and permissions for the KuzApp directory

```
sudo chown -R www-data:www-data /var/www/html/KuzApp
sudo chmod -R 755 /var/www/html/KuzApp
```

STEP 7 --> Configure Database Connection

Create or edit the config.php file in /var/www/html/KuzApp with the correct database credentials

```
<?php
$host = "localhost";
$username = "kuzapp_user";
$password = "your_secure_password";
$dbname = "registration";

$conn = mysqli_connect($host, $username, $password, $dbname);

if (!$conn) {
die("Connection failed: " . mysqli_connect_error());
}
?>
```

STEP 8 --> Verify Background Image

The background image (KuzApp-Fond.jpg) is included in the repository and should be located in /var/www/html/KuzApp.
This image is referenced in the CSS for the application’s background.

STEP 9 --> Test the Application

Open a web browser and navigate to --> https://$IP:8443/KuzApp/login.php

You may need to accept the self-signed certificate warning in your browser.

USAGE INSTRUCTIONS

1. Register a New User

Navigate to the registration page via the "SIGN UP HERE" link on the login page. Enter a username, email, and password, then click "ACTIVATE" to create an account. The password is hashed using SHA-256 before being stored in the database.

2. Log In

On the login page, enter your username and password. If credentials are correct, you will be redirected to app.php, the main dashboard.

3. Use System Administration Tools

The dashboard (app.php) displays a list of buttons for various scripts (e.g., UPDATE-SCRIPT, CHECK-NETWORK). Click a button to execute the corresponding script (via POST requests to respective PHP files). Additional buttons (1–9) allow viewing the script contents in text format before execution.

4. Log Out

Click the "LOG OUT" link in the bottom banner to return to the login page.

SECURITY NOTES

Self-Signed Certificate: The application uses a self-signed SSL certificate, which may trigger browser warnings. For production, consider using a certificate from a trusted Certificate Authority. Password Hashing: Passwords are hashed using SHA-256, but consider upgrading to stronger hashing algorithms like bcrypt in production. Input Sanitization: The application uses mysqli_real_escape_string to prevent SQL injection. Ensure all inputs are validated and sanitized. File Permissions: The /var/www/html/KuzApp directory is owned by www-data with 755 permissions to ensure security.

TROUBLESHOOTING

Apache Errors: Check Apache logs at /var/log/apache2/error.log for issues. Database Connection Issues: Verify config.php credentials and ensure MySQL is running (sudo systemctl status mysql). SSL Issues: Ensure the certificate paths in KuzApp.conf are correct and the SSL module is enabled. Script Execution: Ensure the PHP files for scripts (e.g., update-script.php) exist in the repository and are executable.

FUTUR IMPROVEMENTS

Implement stronger password hashing (e.g., bcrypt). Add input validation for registration and login forms. Support for multi-user roles (e.g., admin vs. regular user). Replace self-signed certificate with a trusted one. Add logging for script execution and user actions.

LICENCE

This project is for educational purposes and is not licensed for commercial use.